Mohamed Essam

AI & Machine Learning Engineer

mohamedessammbuisness@gmail.com 01204668873 linkedin.com/in/mohamed-alattar-2824-
Alexandria - Egypt github.com/MohamedEssam128

PROFESSIONAL SUMMARY

Computer Science Engineer specializing in Artificial Intelligence, Machine Learning, and Deep Learning. Experienced in developing AI solutions using Python, TensorFlow, PyTorch, Computer Vision, NLP, and data analysis, with hands-on experience building end-to-end ML pipelines. Strong problem-solving, analytical thinking, and data-driven decision-making skills with a passion for developing intelligent systems and continuous learning.

EDUCATION

Bachelor of Information Technology 09/2022 – 06/2026
Borg Al - Arab Technological University Alexandria Egypt

PROJECTS

Heritage Preservation AI System | Python, YOLO, Raspberry Pi, Arduino (Graduation Project)
- Designed and developed an AI-powered robotic system using Raspberry Pi and Arduino for heritage digitization, integrating computer vision for automated statue detection and image acquisition with over 95% detection accuracy.
- Generated high-quality 3D reconstructed models from approximately 50 images per statue, achieving accurate geometric reconstruction for digital preservation and hologram fan visualization.
- Built a full-stack heritage management system using MySQL, Firebase, and Flutter for efficient storage, retrieval, and visualization of archaeological data.
- Designed and trained Machine Learning and Deep Learning models for statue classification and damage detection, achieving 94% classification accuracy, 91% mAP@50 for damage detection, and 88% F1-score, with heatmap-based visualization to highlight structural cracks and damaged regions.

Medical Diagnosis NLP System (NHS Data + Transformers + BiLSTM)
- Collected structured medical data from the NHS illness and conditions database for disease classification and symptom analysis.
- Used transformer-based data augmentation to increase dataset diversity by 35% and improve model generalization.
- Developed a BiLSTM model to classify medical conditions from symptoms and causes, achieving 93% accuracy, 91% F1-score, and 92% precision on the validation set.
- Built an end-to-end NLP pipeline including preprocessing, tokenization, feature extraction, and multi-class disease classification, improving model robustness and prediction consistency.

Retail Data Analysis System (SQL + MongoDB + Python + Power BI)
- Designed and implemented a relational SQL database for retail data storage including customers, products, and transactions.
- Used MongoDB for fast data retrieval and flexible querying of semi-structured retail data.
- Extracted and processed data from MongoDB using Python and performed data analysis with Pandas and Matplotlib.
- Built interactive data visualizations and dashboards in Power BI to analyze sales trends, customer behavior, and business performance.
- Built predictive models for sales trend forecasting using Python and Scikit-learn (Linear Regression).

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Java, SQL, C#, Dart, JavaScript
- **AI & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, Keras, Transformers, NLP, Computer Vision, YOLO, OpenCV, Albumentations, LangChain
- **Data Science & Visualization:** Pandas, NumPy, Matplotlib, Power BI, SciPy
- **Databases:** MySQL, PostgreSQL, MongoDB, Firebase
- **Backend & Deployment:** Flask, REST APIs, Docker, AWS
- **Development Tools:** Git, GitHub, Linux, Jupyter Notebook, VS Code
- **Core Computer Science:** Data Structures, Algorithms, Object-Oriented Programming (OOP), Problem Solving

CERTIFICATES

- Andalusia Hospital Internship (AI)

- Networks Fundamentals With Telecom Egypt

COURSES

AI And Data Science Diploma 07/2025 – 04/2026
AMIT Learning